Polymers

Question1

Match List I with List II

List I (Amines)	List II (pK_b)
A. Glyptal	I. Flexible pipes
B. Neoprene	II. Synthetic wool
C. Acrilan	III. Paints and Lacquers
D. LDP	IV. Gaskets

Choose the correct answer from the options given below :- [25-Jan-2023 Shift 2]

Options:

A. A-III, B-II, C-IV, D-I

B. A-III, B-IV, C-II, D-I

C. A-III, B-IV, C-I, D-II

D. A-III, B-I, C-IV, D-II

Answer: B

Question2

Choose the correct answer from the options given below:

List-I	List-II
A. Elastomeric polymer	I. Urea formaldehyderesin
B. Fibre polymer	II. Polystyrene
C. Thermosettingpolymer	III. Polyester
D. Thermoplasticpolymer	IV. Neoprene

[29-Jan-2023 Shift 2]

Options:

A. A-II, B-III, C-I, D-IV

B. A-II, B-I, C-IV, D-III

C. A-IV, B-III, C-I, D-II

D. A-IV, B-I, C-III, D-II

Answer: C

Solution:

Neoprene: Elastomer
Polyester: Fibre
Polystyrene : Thermoplastic
Urea-Formaldhyde Resin: Thermosetting polymer

Question3

Caprolactam when heated at high temperature in presence of water, gives
[30-Jan-2023 Shift 1]

Options:

A. Teflon

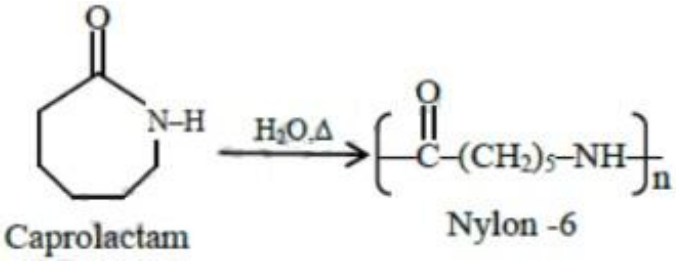
B. Dacron

C. Nylon 6, 6

D. Nylon 6

Answer: D

Solution:



Question4

Polymer used in orlon is:
[6-Apr-2023 shift 1]

Options:

A. Polyacrylonitrile

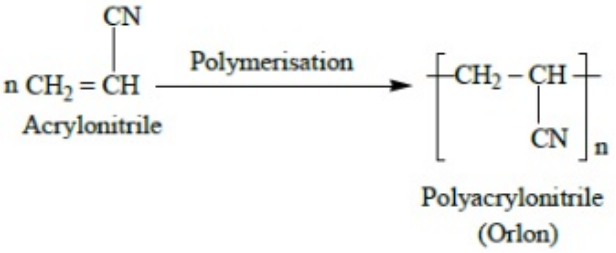
B. Polyethene

C. Polycarbonate

D. Polyamide

Answer: A

Solution:



Question5

Match list I with List II

List I polymer	List II Type/Class
(A) Nylon-2-Nylon-6	(I) Thermosetting polymer
(B) Buna-N	(II) Biodegradable polymer
(C) Urea-Formaldehyde resin	(III) Synthetic rubber
(D) Dacron	(IV) Polyester

Choose the correct answer from the options given below: [10-Apr-2023 shift 1]

Options:

A. (A)-(IV), (B)-(III), (C)-(I), (D)-(II)

B. (A)-(II), (B)-(I), (C)-(IV), (D)-(III)

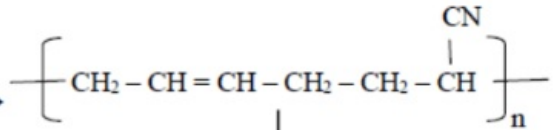
C. (A)-(IV), (B)-(I), (C)-(III), (D)-(II)

D. (A)-(II), (B)-(III), (C)-(I), (D)-(IV)

Answer: D

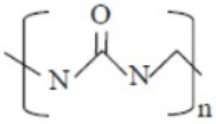
Solution:

Fact Base
(A) Nylon-2-Nylon- 6→ It is α Biodegradable polymer

(B) Buna – N → 

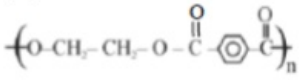
It is a synthetic rubber

(C) Urea – formaldehyde resin



It is a thermos setting polymer

(D) Dacron

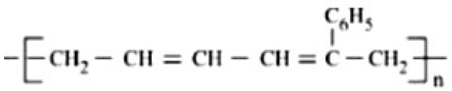
 It is a polyester.

Question6

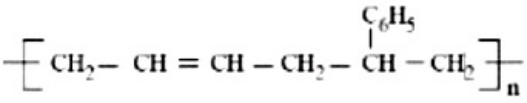
Buna-S can be represented as: [10-Apr-2023 shift 2]

Options:

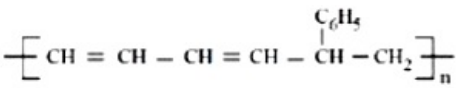
A.



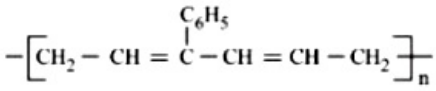
B.



C.

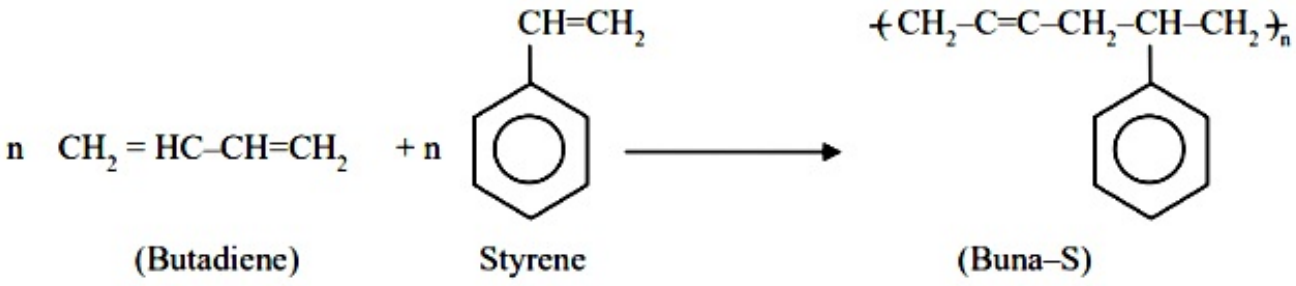


D.



Answer: B

Solution:



Question7

The polymer X-consists of linear molecules and is closely packed. It prepared in the presence of triethylaluminium and titranium tetrachloride under low pressure. The polymer X is-
[11-Apr-2023 shift 1]

Options:

A. Polyacrylonitrile

B. Polytetrafluoroethane

C. High density polythene

D. Low density polythene

Answer: C

Solution:

Solution:
Ethene undergoes addition polymerisation to high density polythene in the presence of catalyst such as AlEt 3 and $TiCl_4$ (Ziegler - Natta catalyst) at a temperature of 333K to 343K and under a pressure of 6 − 7 atmosphere.

Question8

Given below are two statements :
Statement I : Ethene at 333 to 343K and 6-7 atm pressure in the presence of $AlEt_3$ and $TiCl_4$ undergoes addition polymerization to give LDP.
Statement II : Caprolactam at 533-543 K in H_2O through step growth polymerizes to give Nylon 6.
In the light of the above statements, choose the correct answer from the options given below:
[11-Apr-2023 shift 2]

Options:

A. Statement I is true but Statement II is false

B. Both Statement I and Statement II are true

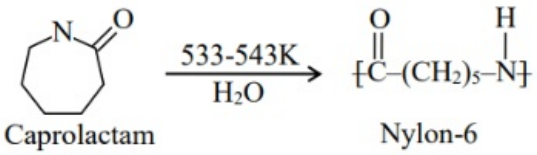
C. Statement I is false but Statement II is true

D. Both Statement I and Statement II are false

Answer: C

Solution:

(Fact-Based)
Statement-I : Ethane at 333 to 343K and 6-7 atm pressure of $AlEt_3$ and $TiCl_4$ undergo addition polymerization to give HDPE not LDPE
Statement-II :



Question9

Match List I with List II

	LIST I (Examples)		LIST II (Type)
A.	2-Chloro-1, 3-butadiene	I.	Biodegradable polymer
B.	Nylon 2-nylone 6	II.	Synthetic Rubber
C.	Polyacrylonitrile	III.	Polyester
D.	Dacron	IV.	Addition Polymer

Choose the correct answer from the options given below: [12-Apr-2023 shift 1]

Options:

A. A-IV, B-I, C-III, D-II

B. A-II, B-IV, C-I, D-III

C. A-II, B-I, C-IV, D-III

D. A-IV, B-III, C-I, D-II

Answer: C

Solution:

Solution:
Fact

Question10

Match The following

Column-A	Column-B
a) Nylon 6	I. Natural Rubber
b) Vulcanized Rubber	II. Cross Linked
c) cis-1, 4-polyisoprene	III. Caprolactam
d) Polychloroprene	IV. Neoprene

Choose the correct answer from options given below: [13-Apr-2023 shift 1]

Options:

A. a → II, a → III, c → IV , d → I

B. a → IV, b → III, c → II, d → I

C. a→ III, b → II, c → I, d → IV

D. a → III, b → IV, c → I, d → II

Answer: B

Solution:

Solution:
Nylon-6 - Caprolactum (Monomer)
Natural rubber- Isoprene (Monomer)
Vulcanized rubber - Sulphur containing rubber
Neoprene- Chloroprene (Monomer)

Question11

Match List-I with List-II :

List-I (Monomer)	List-II (Polymer)
(A) Tetrafluoroethene	(I) Orlon
(B) Acrylonitrile	(II) Natural rubber
(C) Caprolactam	(III) Teflon
(D) Isoprene	(IV) Nylon-6

Choose the correct answer from the options given below : [15-Apr-2023 shift 1]

Options:

A. (A)-(II), (B)-(III), (C)-(IV), (D)-(I)

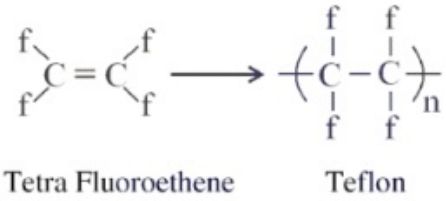
B. (A)-(III), (B)-(I), (C)-(IV), (D)-(II)

C. (A)-(IV), (B)-(I), (C)-(II), (D)-(III)

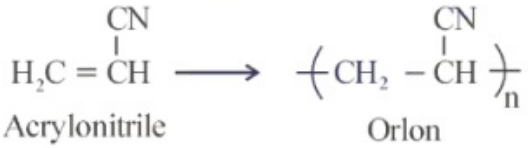
D. (A)-(III), (B)-(IV), (C)-(II), (D)-(I)

Answer: B

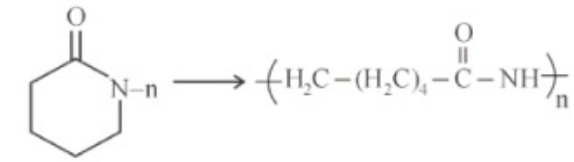
Solution:



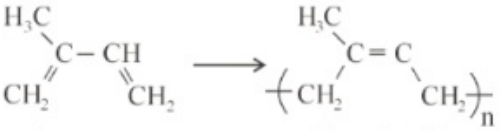
Tetra Fluoroethene Teflon



Acrylonitrile Orlon



Caprolactum Nylon - 6



Isoprene Natural rubber

Question12

Which of the following is an example of polyester? [24-Jun-2022-Shift-1]

Options:

A. Butadiene-styrene copolymer

B. Melamine polymer

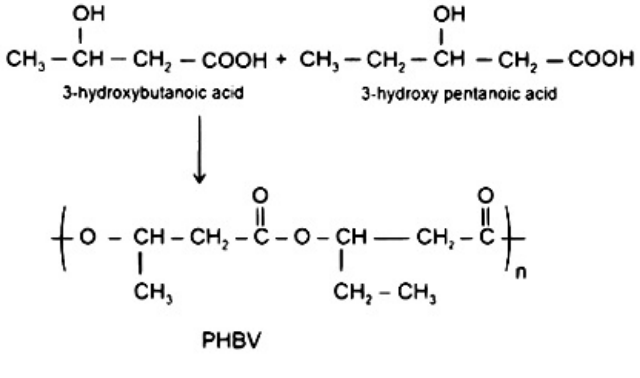
C. Neoprene

D. Poly- β-hydroxybutyrate-co- β-hydroxyvalerate

Answer: D

Solution:

Solution:
Polyesters are formed by condensation reaction between alcohols and carboxylic acid.
Poly- β-hydroxybutyrate-co- β-hydroxy valerate (PHBV) is a polymer obtained by condensation reaction of 3 hydroxy butanoic acid with 3-hydroxypentanoic acid.



Question13

Which of the following is not an example of a condensation polymer? [24-Jun-2022-Shift-2]

Options:

A. Nylon 6,6

B. Decron

C. Buna-N

D. Silicone

Answer: C

Solution:

Solution:
Nylon 6,6 is a condensation polymer of hexamethylene diamine and adipic acid
Dacron is a condensation polymer of terephthalic acid and ethylene glycol.
Buna- N is an addition polymer of 1,3 -butadiene and acrylonitrile
Silicone is a condensation polymer of dialkyl silanediol.

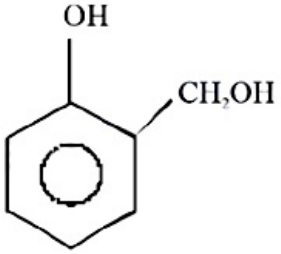
Question14

The Novolac polymer has mass of 963g. The number of monomer units present in it are [25-Jun-2022-Shift-2]

Answer: 9

Solution:

Solution:
Monomer unit of Novolac is



its molecular mass is 124 amu.
Upon considering molecular weight of polymer as 963 amu (In question its given as 963 gram) Now if during formation of Novolac, $(n-1)$ unit of water are removed then
$n \times 124 = 963 + [18 \times (n-1)]$
$n = 9$

Question15

Which of the following sets are correct regarding polymer?
(A) Copolymer : Buna-S
(B) Condensation polymer : Nylon-6,6
(C) Fibres : Nylon-6,6
(D) Thermosetting polymer : Terylene
(E) Homopolymer : Buna-N
Choose the correct answer from given options below :
[26-Jun-2022-Shift-1]

Options:

A. (A), (B) and (C) are correct

B. (B), (C) and (D) are correct

C. (A), (C) and (E) are correct

D. (A), (B) and (D) are correct

Answer: A

Solution:

Solution:
(A) Buna-S − Copolymer
(B) Nylon-6,6 − Condensation polymer
(C) Nylon-6,6 - Fibre
(D) Terylene − Thermoplastic
(E) Buna-N - Copolymer
∴A, B and C are correct.

Question16

Choose the correct answer from the options given below:

	List - I (Polymer)		List - II (Used in)
(A)	Bakelite	(I)	Radio and television cabinets
(B)	Glyptal	(II)	Electrical switches
(C)	PVC	(III)	Paints and Lacquers
(D)	Polystyrene	(IV)	Water pipes

[27-Jun-2022-Shift-1]

Options:

A. (A) - (II), (B) - (III), (C) - (IV), (D) - (I)

B. (A) - (I), (B) - (II), (C) - (III), (D) - (IV)

C. (A) - (IV), (B) - (III), (C) - (II), (D) - (I)

D. (A) − (II), (B) − (III), (C) − (I), (D) − (IV)

Answer: A

Solution:

Solution:

	List - I (Polymer)		List - II (Used in)
(A)	Bakelite	(II)	Electrical switches
(B)	Glyptal	(III)	Paints and Lacqures
(C)	PVC	(IV)	Water pipes
(D)	Polystyrene	(I)	Radio and television Cabinets

Question17

Which is true about Buna-N? [27-Jun-2022-Shift-2]

Options:

A. It is a linear polymer of 1, 3-butadiene.

B. It is obtained by copolymerization of 1, 3-butadiene and styrene.

C. It is obtained by copolymerization of 1, 3-butadiene and acrylonitrile.

D. The suffix N in Buna-N stands for its natural occurrence.

Answer: C

Solution:

Solution:
It is copolymerization of 1, 3-butadiene and acrylonitrile.

Question18

Which one of the following is NOT a copolymer? [28-Jun-2022-Shift-1]

Options:

A. Buna-S

B. Neoprene

C. PHBV

D. Butadiene-styrene

Answer: B

Solution:

Solution:
Buna-S, PHBr and Butadiene-styrene are copolymer. Only neoprene is namopolymer.

Question19

Given below are two statements, one is Assertion (A) and other is Reason (R).
Assertion (A) : Natural rubber is a linear polymer of isoprene called sis-polyisoprene with elastic properties.
Reason (R) : The cis-polyisoprene molecules consist of various chains held together by strong polar interactions with coiled structure.
In the light of the above statements, choose the correct one from the options given below :
[28-Jun-2022-Shift-2]

Options:

A. Both (A) and (R) are true and (R) is the correct explanation of (A).

B. Both (A) and (R) are true but (R) is not the correct explanation of (A).

C. (A) is true but (R) is false.

D. (A) is false but (R) is true.

Answer: C

Solution:

Solution:
Natural rubber is linear polymer of isoprene (2- methyl-1,3-butadiene) and is also called cis-1,4polyisoprene. The cis-polyisoprene molecules consists of various chains held together by weak Vander Waal"s interactions and has a coiled structure.

Question20

The polymer, which can be stretched and remains its original status on releasing the force is
[29-Jun-2022-Shift-1]

Options:

A. Bakelite.

B. Nylon 6, 6.

C. Buna-N.

D. Terylene.

Answer: C

Solution:

Solution:
Buna – N is synthetic rubber which can be stretched and retains its original status on releasing the force.

Question21

Given below are two statements : One is labelled as Assertion A and the other is labelled as Reason R.
Assertion A : Dacron is an example of polyester polymer.
Reason R : Dacron is made up of ethylene glycol and terephthalic acid monomers.
In the light of the above statements, choose the most appropriate answer from the options given below.
[29-Jun-2022-Shift-2]

Options:

A. Both A and R are correct and R is the correct explanation of A.

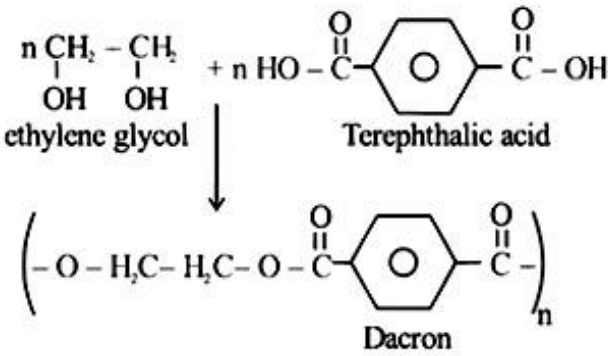
B. Both A and R are correct but R is NOT the correct explanation of A.

C. A is correct but R is not correct.

D. A is not correct but R is correct.

Answer: A

Solution:

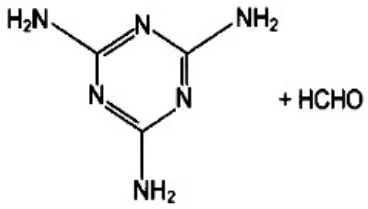


Question22

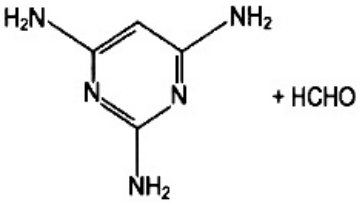
Melamine polymer is formed by the condensation of:
[25-Jul-2022-Shift-1]

Options:

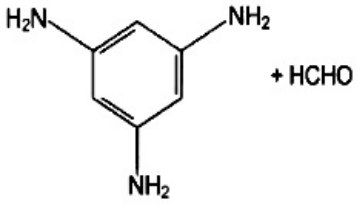
A.



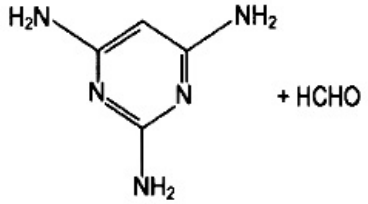
B.



C.



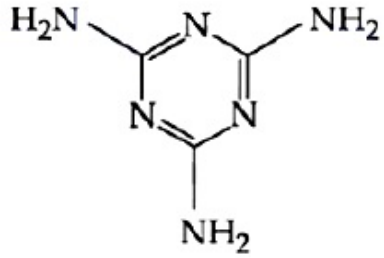
D.



Answer: A

Solution:

Melamine :



Formaldehyde HCHO
Melamine formaldehyde Resin is melamine polymer

Question23

List I		List II	
(A)	Nylon 6,6	(I)	Buckets
(B)	Low density polythene	(II)	Non-stick utensils
(C)	High density polythene	(III)	Bristles of brushes
(D)	Teflon	(IV)	Toys

Choose the correct answer from the options given below : [25-Jul-2022-Shift-2]

Options:

A. A-III, B-I, C-IV, D-II

B. A-III, B-IV, C-I, D-II

C. A-II, B-I, C-IV, D-III

D. A-II, B-IV, C-I, D-III

Answer: B

Solution:

Solution:
Nylon 6, 6→ used in making bristles of brushes
Low density polythene → used in making Toys
High density polythene → used in making Buckets
Teflon → used in making non-stick utensils

Question24

Vulcanization of rubber is carried out by heating a mixture of [26-Jul-2022-Shift-2]

Options:

A. isoprene and styrene

B. neoprene and sulphur

C. isoprene and sulphur

D. neoprene and styrene

Answer: C

Solution:

Solution:
When a mixture of isoprene and sulphur is heated, isoprene gets polymerised to natural rubber and then vulcanization of natural rubber with sulphur takes place.

Question25

Match List I with List II.

List I Polymers	List II Commercial names
(A) Phenol-formaldehyde resin	(I)Glyptal
(B) Copolymer of 1,3-butadiene and sterene	(II) Novolac
(C) Polyester of glycol and phthalic acid	(III) Buna-S
(D) Polyester of glycol and terephthalic acid	(IV) Dacron

Choose the correct answer from the options given below: [27-Jul-2022-Shift-1]

Options:

A. A-II, B-III, C-IV, D-I

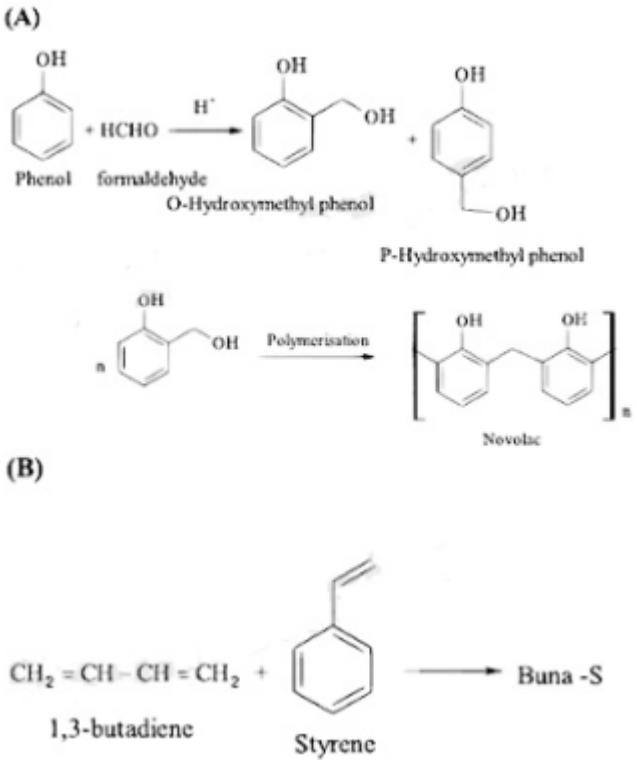
B. A-II, B-III, C-I, D-IV

C. A-II, B-I, C-III, D-IV

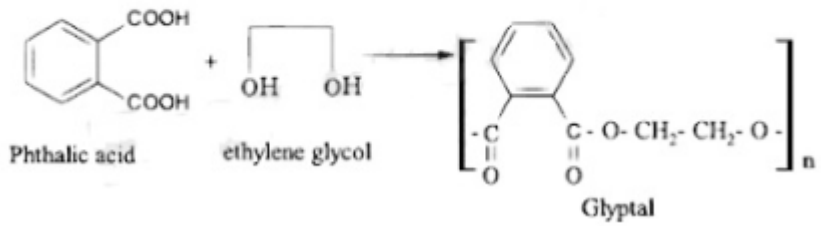
D. A-III, B-II, C-IV, D-I

Answer: B

Solution:

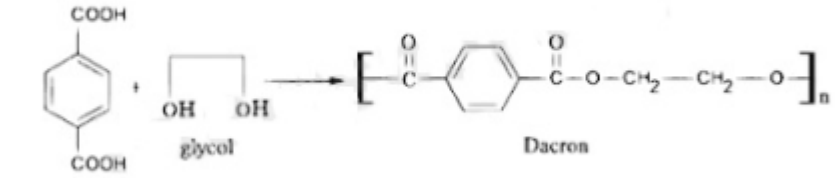


(C)



Glyptal

(D)



Terephthalic acid

Dacron

Question26

Match List - I with List - II.

List I (Polymer)	List II (Monomer)
(A)Neoprene	(I)Acrylonitrile
(B)Teflon	(II)Chloroprene
(C)Acrilan	(III)Tetrafluoroethene
(D)Natural rubber	(IV)Isoprene

Choose the correct answer from the options given below : [27-Jul-2022-Shift-2]

Options:

A. (A) − (II), (B) − (III), (C) − (I), (D) − (IV)

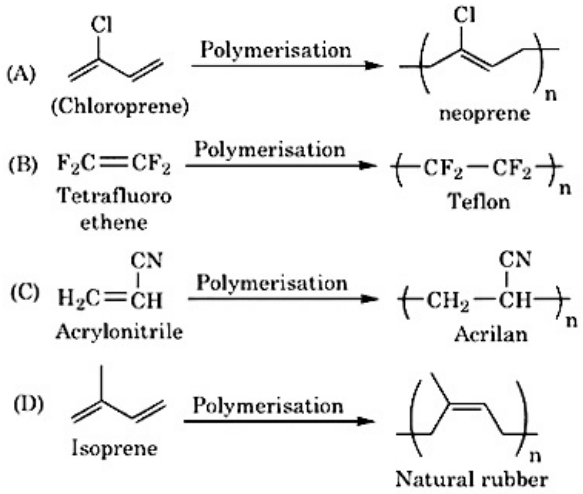
B. (A) − (II), (B) − (I), (C) − (III), (D) − (IV)

C. (A) − (II), (B) − (I), (C) − (IV), (D) − (III)

D. (A) − (I), (B) − (II), (C) − (III), (D) − (IV)

Answer: A

Solution:



Question27

Terylene polymer is obtained by condensation of : [28-Jul-2022-Shift-1]

Options:

A. Ethane-1,2-diol and Benzene-1,3 dicarboxylic acid

B. Propane-1, 2-diol and Benzene-1, 4 dicarboxylic acid

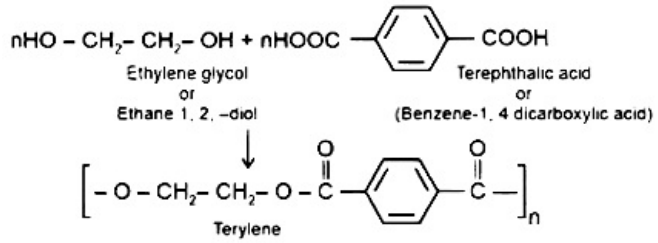
C. Ethane-1,2-diol and Benzene-1, 4 dicarboxylic acid

D. Ethane-1,2-diol and Benzene-1, 2 dicarboxylic acid

Answer: C

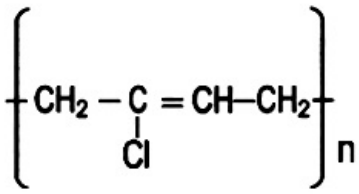
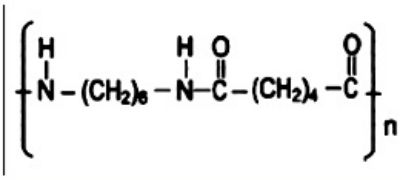
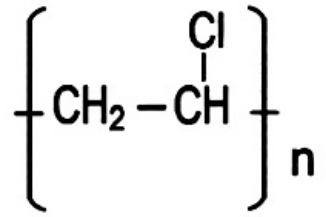
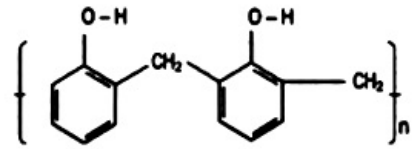
Solution:

Solution:



Question28

Match List I with List II

List - I (Polymer)	List - II (Nature)
	(I)Thermosetting polymer
	(II)Fibers
	(III)Elastomer
	(IV)Thermoplastic polymer

Choose the correct answer from the options given below : [28-Jul-2022-Shift-2]

Options:

A. A-II, B-III, C-IV, D-I

B. A-III, B-II, C-IV, D-I

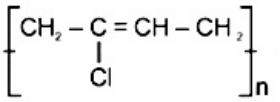
C. A-III, B-I, C-IV, D-II

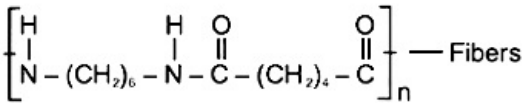
D. A-I, B-III, C-IV, D-II

Answer: B

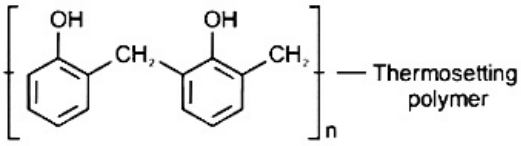
Solution:

Solution:

(A)  — Elastomer

(B)  — Fibers

(C)  — Thermoplastic polymer

(D)  — Thermosetting polymer

---

Question29

Which of the following is NOT a natural polymer? [29-Jul-2022-Shift-2]

Options:

A. Protein

B. Starch

C. Rubber

D. Rayon

Answer: D

Solution:

Solution:
Starch, Protein, Rubber are natural polymer.
Rayon is a semi-synthetic fiber, made from natural sources of regenerated cellulose, such as wood and related agricultural products.

Question30

The structure of neoprene is [26 Feb 2021 Shift 1]

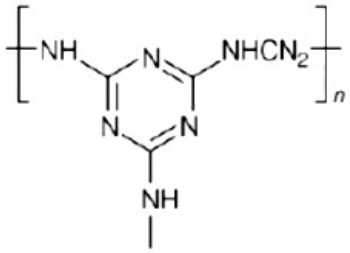
Options:

A. $\left[CH_2CH = CH - CH_2 - CH_2 - \underset{N}{\overset{|}{CH}} \right]_n$

B. $\left[CH_2 - \underset{CN}{\overset{|}{CH}} \right]_n$

C. $\left[CH_2 - \overset{Cl}{\underset{|}{C}} = CH - CH_2 \right]_n$

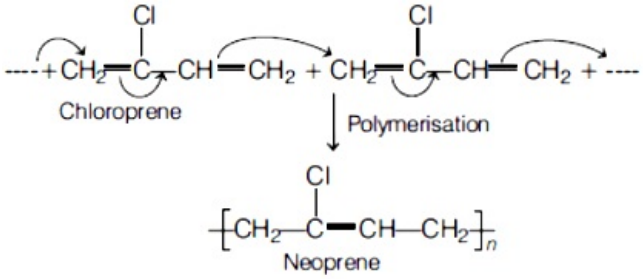
D.



Answer: C

Solution:

Solution:
Neoprene is an addition homopolymer of chloroprene.



Question31

Which statement is correct?
[25 Feb 2021 Shift 1]

Options:

A. Buna- N is a natural polymer.

B. Buna- S is a synthetic and linear thermosetting polymer.

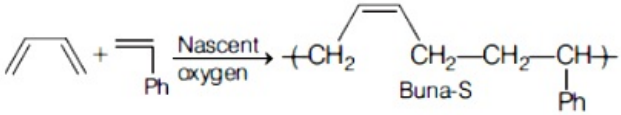
C. Neoprene is an addition copolymer used in plastic bucket manufacturing.

D. Synthesis of buna- S needs nascent oxygen.

Answer: D

Solution:

Only statement (d) is correct whereas other statements are incorrect. The correct statements are as follows
Buna- N is a synthetic polymer.
Buna- S is an elastomer. Buna- S is polymerised by addition polymerisation method, which needs radical initiator for chain propagation step. Nascent oxygen can be used as an radical initiator.



Neoprene is a synthetic rubber not an addition copolymer.

Question32

In polymer buna-S: ' S ' stands for
[24 Feb 2021 Shift 2]

Options:

A. sulphonation

B. strength

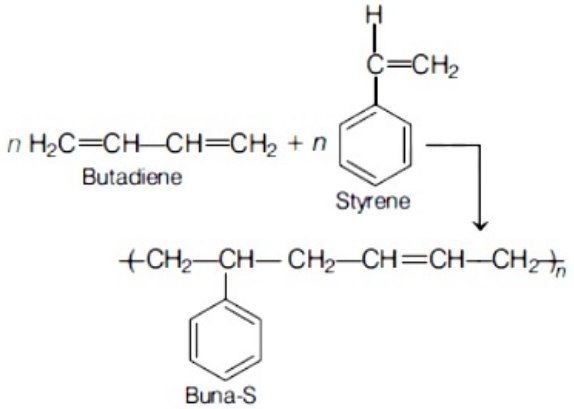
C. sulphur

D. styrene

Answer: D

Solution:

Solution:
Buna-S is the co-polymer of but 1,3 -diene and styrene as follows.



Styrene butadiene rubber is also called buna- S, in which ' Bu ' stands for butadiene, 'na' stands for sodium and ' S ' stands for styrene.
Since, butadiene and styrene is one of the constituent monomer of given polymer.

Question33

Match List-I with List-II.

List-I (Monomer Unit)	List-II (Polymer)
(A) Caprolactum	(i) Natural rubber
(B) 2-Chloro-1, 3-butadiene	(ii) Buna-N
(C) Isoperene	(iii) Nylon 6
(D) Acrylonitrile	(iv) Neoprene

Choose the correct answer from the options given below:
[24 Feb 2021 Shift 1]

Options:

A. (A)-(iv), (B)-(iii), (C)-(ii), (D)-(i)

B. (A)-(ii), (B)-(i), (C)-(iv), (D)-(iii)

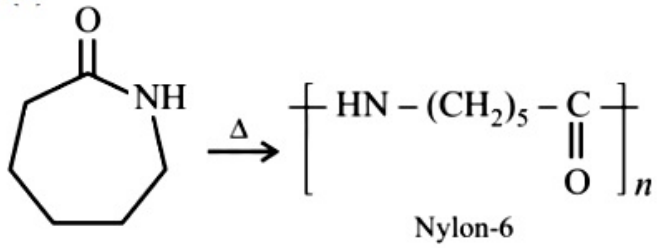
C. (A)-(iii), (B)-(iv), (C)-(i), (D)-(ii)

D. (A)-(i), (B)-(ii), (C)-(iii), (D)-(iv)

Answer: C

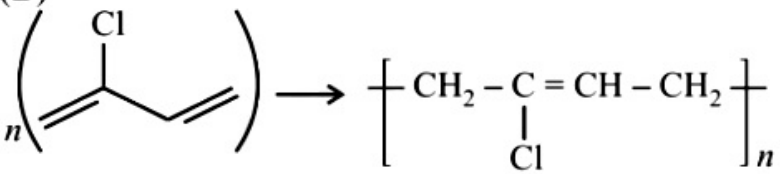
Solution:

(A)



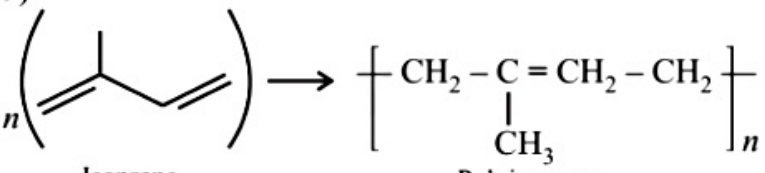
Nylon-6

(B)



2-chloro-1, 3-butadiene Neoprene (synthetic rubber)

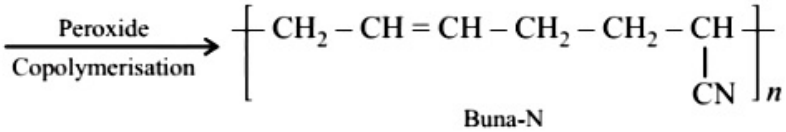
(C)



Isoprene (2-Methylbuta-1, 3-diene) Polyisoprene (Natural rubber)

(D) $n CH_2 = CH - CH = CH_2 + CH_2 = CH - CN$

1,3-Butadiene Aerylonitrile



Buna-N

Question34

Match List-I with List-II.

List-I (Chemicals)	List-II (Use / Preparation/Constituent)
(A) Alcoholic potassium hydroxide	(i) Electrodes in batteries
(B) $Pd/BaSO_4$	(ii) Obtained by addition reaction
(C) BHC (Benzene hexachloride)	(iii) Used for β-elimination reaction
(D) Polyacetylene	(iv) Lindlar catalyst

Choose the most appropriate match [18 Mar 2021 Shift 1]

Options:

A. A - 2, B - 1, C - 4, D - 3

B. A - 3, B - 4, C - 2, D - 1

C. A - 3, B - 1, C - 4, D - 2

D. A - 2, B - 4, C - 1, D - 3

Answer: B

Solution:

Solution:
A. Alcoholic KOH is used for β - elimination reaction. It will form alkane as a major product.
B. $Pd/BaSO_4$ (Lindlar's catalyst) is used for convert alkyne to alkane.
C. BHC (Benzene hexachloride) is used to obtained addition product of benzene and chlorine.
D. Polyacetylene is used as electrodes in batteries.
∴ Correct match is (A-iii), (B-iv), (C-ii), (D-i).

Question35

Which of the following polymer is used in the manufacture of wood laminates ?
[16 Mar 2021 Shift 2]

Options:

A. cis-polyisoprene

B. Melamine formaldehyde resin

C. Urea formaldehyde resin

D. Phenol and formaldehyde resin

Answer: C

Solution:

Solution:
(c) Urea-formaldehyde is used to manufacture unbreakable cups and laminated sheets. It is an example of condensation polymer in which water molecule is removed from urea and formaldehyde forming an amide linkage.
(a) cis-poly isoprene is natural rubber.
(b) Melamine - formaldehyde polymer used in the manufacture of unbreakable crockery.
(d) Phenol-formaldehyde resin used as good wood adhesives for plywood and particle board.

Question36

Staggered and eclipsed conformers of ethane are :
[27 Jul 2021 Shift 1]

Options:

A. Polymers

B. Rotamers

C. Enantiomers

D. Mirror images

Answer: B

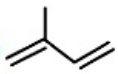
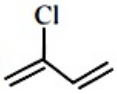
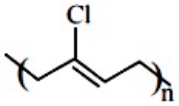
Solution:

Staggered and eclipsed conformers of ethane also known as rotamers

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Question37

Match List-I with List-II :

List-I	List-II
(a) Chloroprene	(i) 
(b) Neoprene	(ii) 
(c) Acrylonitrile	(iii) 
(d) Isoprene	(iv) $CH_2 = CH-CN$

Choose the correct answer from the options given below : [22 Jul 2021 Shift 2]

Options:

A. (a) - (iii), (b)-(iv), (c) -(ii), (d) -(i)

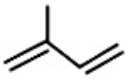
B. (a) - (ii), (b)-(iii), (c) -(iv), (d) -(i)

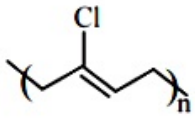
C. (a) - (ii), (b)-(i), (c) -(iv), (d) -(iii)

D. (a) - (iii), (b)-(i), (c) -(iv), (d) -(ii)

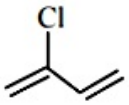
Answer: B

Solution:

(a) Chloroprene (i) 

(b) Neoprene (ii) 

(c) Acrylonitrile (iii) CH_2=CH–CN

(d) Isoprene (iv) 

Question38

Orlon fibres are made up of :
[20 Jul 2021 Shift 1]

Options:

A. Polyacrylonitrile

B. Polyesters

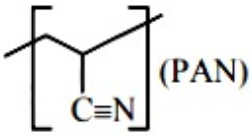
C. Polyamide

D. Cellulose

Answer: A

Solution:

→ orlon fibers are made up of Polyacrylonitrile



Question39

A biodegradable polyamide can be made from:
[25 Jul 2021 Shift 2]

Options:

A. Glycine and isoprene

B. Hexamethylene diamine and adipic acid

C. Glycine and aminocaproic acid

D. Styrene and caproic acid

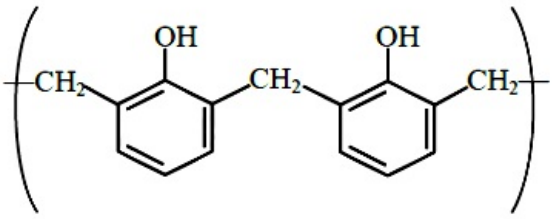
Answer: C

Solution:

Solution:
A biodegradable polyamide nylon-2-Nylon-6 in made from glycine and amino caproic acid

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Question40



is a repeating unit for :
[25 Jul 2021 Shift 1]

Options:

A. Novolac

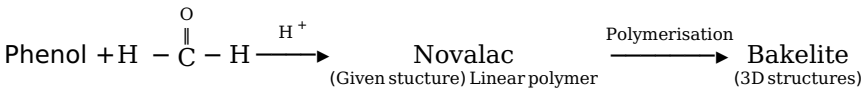
B. Buna–N

C. Acrilan

D. Neoprene

Answer: A

Solution:



Novolac is low molecular weight polymer derived from phenol and formaldehyde. They are related to bakelite, which is more highly cross-linked.

Question41

Bakelite is a cross-linked polymer of formaldehyde and :
[20 Jul 2021 Shift 2]

Options:

A. PHBV

B. Buna-S

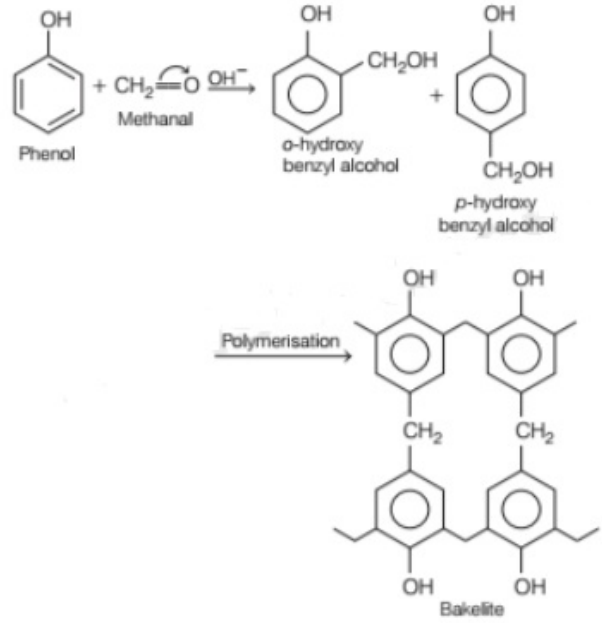
C. Novolac

D. Dacron

Answer: C

Solution:

Bakelite is a cross-linked polymer of formaldehyde and novolac.



Question42

The polymer formed on heating novolac with formaldehyde is [26 Aug 2021 Shift 1]

Options:

A. bakelite

B. polyester

C. melamine

D. nylon 6,6

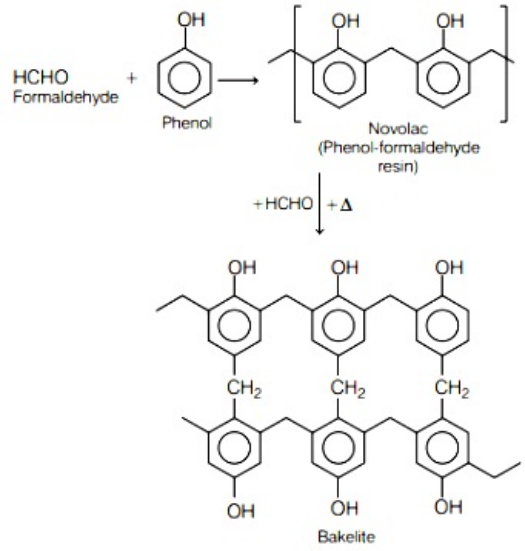
Answer: A

Solution:

Solution:

$$\text{Novolac} + \text{Formaldehyde} \xrightarrow{\Delta} \text{Bakelite}$$

Novolac when heated with formaldehyde will undergoes cross linking to form an infusible solid mass called bakelite.



Question43

Which among the following is not a polyester ? [31 Aug 2021 Shift 2]

Options:

A. Novolac

B. PHBV

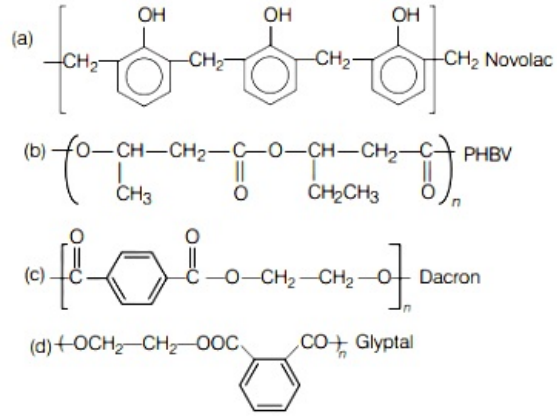
C. Dacron

D. Glyptal

Answer: A

Solution:

Solution:
The structure of the given polymers are as follows



∵ Novolac does not contain any ester group.
∴ It is not a polyester.

Question44

Monomer of novolac is
[31 Aug 2021 Shift 1]

Options:

A. 3-hydroxybutanoic acid

B. phenol and melamine

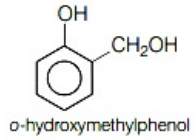
C. o-hydroxymethylphenol

D. 1,3-butadiene and styrene

Answer: C

Solution:

Solution:
The monomer of novolac is o-hydroxymethylphenol.



o-hydroxymethylphenol

Hence, correct option is (c).

Question45

Monomer units of dacron polymer are
[1 Sep 2021 Shift 2]

Options:

A. ethylene glycol and phthalic acid

B. ethylene glycol and terephthalic acid

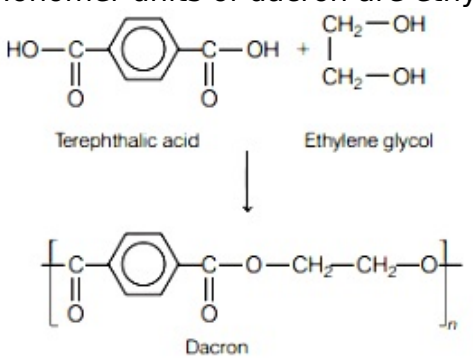
C. glycerol and terephthalic acid

D. glycerol and phthalic acid

Answer: B

Solution:

Monomer units of dacron are ethylene glycol and terephthalic acid.



Question46

Which polymer has 'chiral' monomer(s)? [Jan. 09,2020(I)]

Options:

A. Neoprene

B. Buna-N

C. N yl on6, 6

D. PHBV

Answer: D

Solution:

Solution:

$CH_3 - \underset{(3-\text{Hydroxybutanoic acid})}{CH} - CH_2 - COOH$ and $CH_3 - \underset{(3-\text{Hydroxypentanoic acid})}{CH_2} - CH^{-CH_2}COOH$

Both monomers of PHBV have chiral centre.

Question47

Preparation of Bakelite proceeds via reactions: [Jan. 08,2020 (II)]

Options:

A. Electrophilic addition and dehydration

B. Condensation and elimination

C. Electrophilic substitution and dehydration

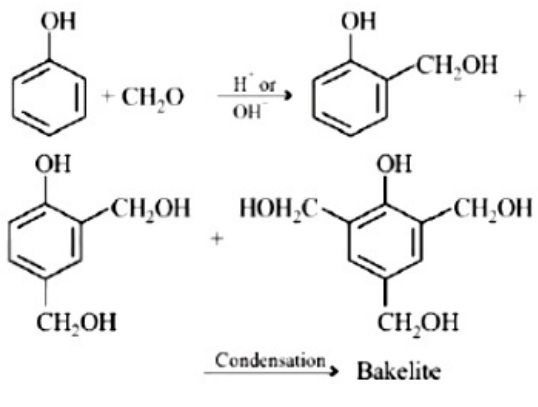
D. Nucleophilic addition and dehydration

Answer: C

Solution:

Solution:

Formation of Bakelite follows electrophilic substitution reaction of phenol with formaldehyde followed by dehydration.



Question48

Which one of the following polymers is not obtained by condensation polymerisation?
[Sep. 05, 2020 (II)]

Options:

A. Nylon 6,6

B. Buna - N

C. Bakelite

D. Nylon 6

Answer: B

Solution:

Solution:
Buna- N is obtained by addition polymerisation.

$$\underset{1,3-\text{Butadicne}}{nCH_2=CH-CH=CH_2}+\underset{\text{Acrylonitrile}}{nCH_2=CH}\xrightarrow{\text{copolymensation}}\underset{\text{Buna}-N}{(CH_2-CH=CH-CH_2-CH_2-CH)_n}$$

Nylon-6, bakelite, nylon-6. 6 are obtained by condensation polymerisation.

Question49

Consider the Assertion and Reason given below.
Assertion (A) : Ethene polymerized in the presence of Ziegler Natta Catalyst at high temperature and pressure is used to make buckets and dustbins.
Reason (R) : High density polymers are closely packed and are chemically inert.
Choose the correct answer from the following:

[Sep. 06, 2020(I)]

Options:

A. (A) is correct but (R) is wrong.

B. Both (A) and (R) are correct but (R) is not the correct explanation of (A).

C. Both (A) and (R) are correct but (R) is the correct explanation of (A).

D. (A) and (R) both are wrong.

Answer: C

Solution:

High density polythene is formed when addition polymerisation of ethene takes place in the presence of a catalyst (Ziegler Natta catalyst) at a high temp. ($333K$ to $349K$) and under low pressure ($6-7atm$). High density polyethene is hard and chemically inert due to close packing thats why used to make buckets and dustbins.

Question50

The correct match between Item-I and Item - II is:

Item-I	Item-II
(a) Natural rubber	(I) 1,3 -butadiene + styrene
(b) Neoprene	(II) 1,3 -butadiene+ acrylonitrile
(c) Buna-N	(III) Chloroprene
(d) Buna-S	(IV) Isoprene

[Sep.06,2020(II)]

Options:

A. $(A)-(III),(B)-(IV),(C)-(I),(D)-(II)$

B. $(A)-(III),(B)-(IV),(C)-(II),(D)-(T)$

C. $(A)-(IV),(B)-(III),(C)-(II),(D)-(I)$

D. $(A)-(IV),(B)-(III),(C)-(I),(D)-(II)$

Answer: C

Solution:

Solution:
(A) Natural rubber - Polymer of isoprene
(B) Neoprene - Polymer of chloroprene
(C) Buna N - Polymer of 1,3 -butadiene and acrylonitrile
(D) Buna S - Polymer of 1,3 -butadiene and styrene

Question51

Poly- β -hydroxybutyrate-co- β -hydroxy valerate (PH BV) is a copolymer of _____.
[Jan. 12, 2019 (I)]

Options:

A. 3-Hydroxybutanoic acid and 4-Hydroxypentanoic acid

B. 2-Hydroxybutanoic acid and 3-Hydroxypentanoic acid

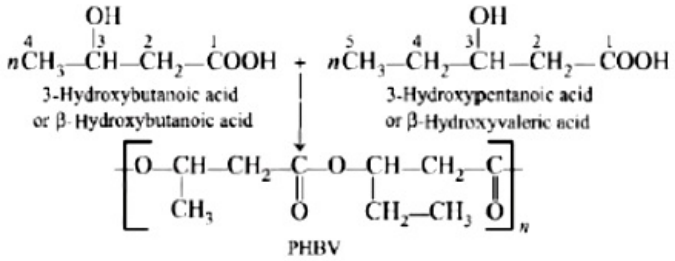
C. 3-Hydroxybutanoic acid and 2-Hydroxypentanoic acid

D. 3-Hydroxybutanoic acid and 3-Hydroxypentanoic acid

Answer: D

Solution:

Solution:
As the name of the polymer indicates, it is a copolymer of following two monomers.



Question52

The two monomers for the synthesis of nylon 6,6 are : [Jan. 12,2019 (II)]

Options:

A. $HOOC(CH_2)_4COOH$, $H_2N(CH_2)_6NH_2$

B. $HOOC(CH_2)_6COOH$, $H_2N(CH_2)_6NH_2$

C. $HOOC(CH_2)_4COOH$, $H_2N(CH_2)_4NH_2$

D. $HOOC(CH_2)_6COOH$, $H_2N(CH_2)_4NH_2$

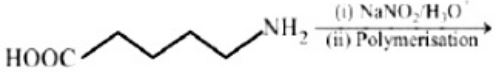
Answer: A

Solution:

Solution:
\ Adipic acid (having 6C) and hexamethylene diamine (having 6C).

Question53

The polymer obtained from the following reactions is



[Jan. 11,2019(I)]

Options:

A. $\left[\overset{O}{\overset{\|}{C}} - (CH_2)_4 - \overset{H}{N} \right]_n$

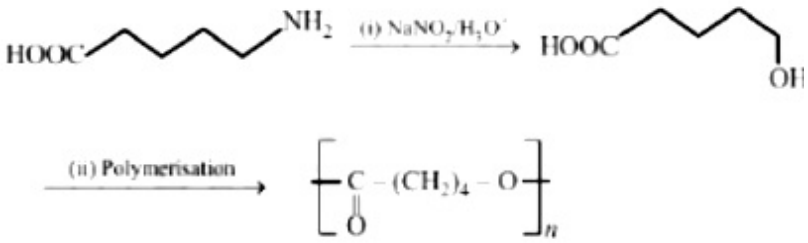
B. $\left[HN - \overset{O}{\overset{\|}{C}} - (CH_2)_4 - \overset{O}{\overset{\|}{C}} - \overset{H}{N} \right]_n$

C. $\left[O-(CH_2)_4-\overset{O}{\overset{\|}{C}} \right]_n$

D. $\left[O-\overset{O}{\overset{\|}{C}}-(CH_2)_4-O \right]_n$

Answer: C

Solution:



Question54

The homopolymer formed from 4 -hydroxy-butanoic acids is : [Jan. 11, 2019 (II)]

Options:

A. $\left[O-\overset{O}{\overset{\|}{C}}-(CH_2)_3-O \right]_n$

B. $\left[O-\overset{O}{\overset{\|}{C}}(CH_2)_2-\overset{O}{\overset{\|}{C}} \right]_n$

C. $\left[O-\overset{O}{\overset{\|}{C}}-(CH_2)_2-\overset{O}{\overset{\|}{C}}-O \right]_n$

D. $\left[\overset{O}{\overset{\|}{C}}(CH_2)_3-O \right]_n$

Answer: D

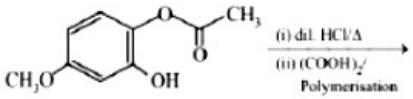
Solution:

Solution:

$\underset{\text{4 Hydroxybutanoic acid}}{\overset{C}{\underset{OH}{|}}}H_2-CH_2-CH_2-\overset{O}{\overset{\|}{C}}-OH \xrightarrow{\text{Polymerisation}} \underset{\text{Poly – 4 hydroxybutyrate}}{\left[O-(CH_2)_3-\overset{O}{\overset{\|}{C}} \right]_n}$

Question55

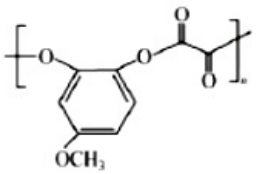
The major product of the following reaction is:



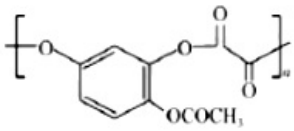
[Jan. 10,2019 (II)]

Options:

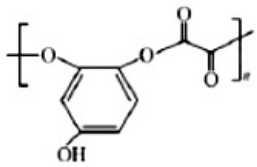
A.



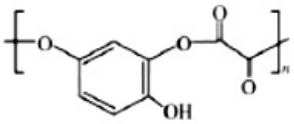
B.



C.



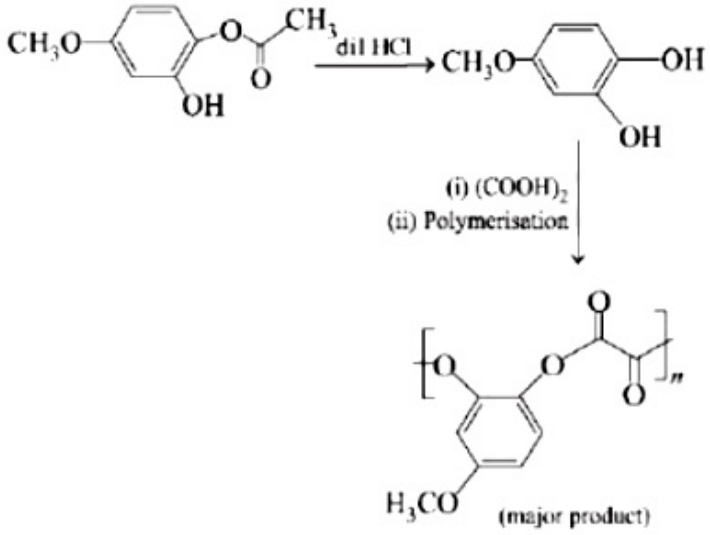
D.



Answer: A

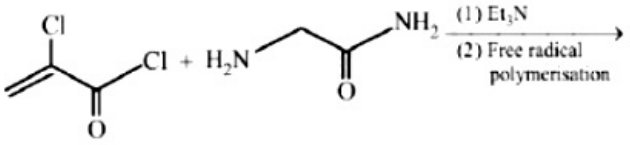
Solution:

Reaction involved:



Question56

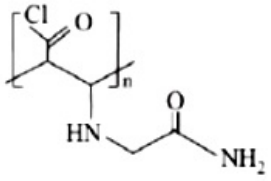
Major product of the following reaction is:



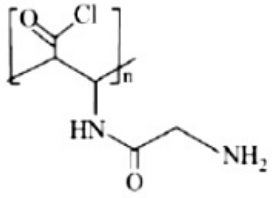
[Jan. 9, 2019(I)]

Options:

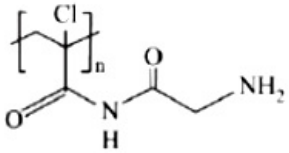
A.



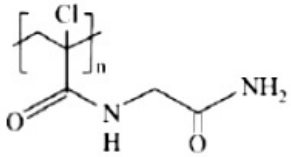
B.



C.



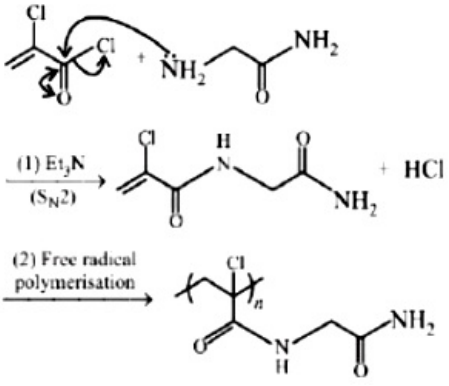
D.



Answer: D

Solution:

Mechanism for the formation of major product is as follows:



Question57

Which of the following is a thermosetting polymer? [April 12, 2019 (I)]

Options:

A. Bakelite

B. Buns-N

C. Nylon 6

D. PVC

Answer: A

Solution:

Solution:
Bakelite is an example of thermosetting polymer. PVC and Nylon 6 are thermoplastic and Buna-N is an elastomer.

Question58

The correct name of the following polymer is:



[April 21, 2019(II)]

Options:

A. Polyisobutane

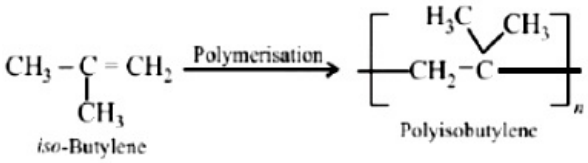
B. Polytert-butylene

C. Polyisoprene

D. Polyisobutylene

Answer: D

Solution:



Question59

Which of the following is a condensation polymer? [April 10, 2019 (I)]

Options:

A. Buna - S

B. Neoprene

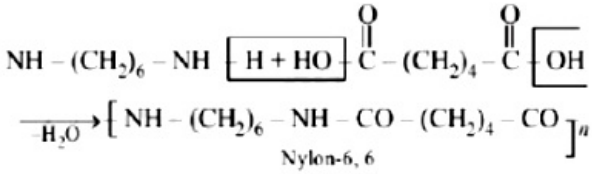
C. Teflon

D. Nylon 6,6

Answer: D

Solution:

Solution:
Nylon −6, 6 is obtained by condensation polymerisation of hexamethylene diamine and adipic acid.



So, nylon- 6,6 is a condensation polymer. Other given polymers are addition polymers.

Question60

The structure of Nylon − 6 is: [April 10, 2019 (I)]

Options:

A. $[(CH_2)_4 - \overset{O}{\overset{\|}{C}} - \overset{H}{\overset{|}{N}}]_n$

B. $[\overset{O}{\overset{\|}{C}} - (CH_2)_6 - \overset{H}{\overset{|}{N}}]_n$

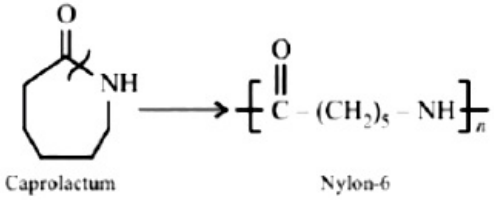
C. $[\overset{O}{\overset{\|}{C}} - (CH_2)_5 - \overset{H}{\overset{|}{N}}]_n$

D. $[(CH_2)_6 - \overset{O}{\overset{\|}{C}} - \overset{H}{\overset{|}{N}}]_n$

Answer: C

Solution:

Solution:
Nylon- 6 is prepared from caprolactam.



Question61

The correct match between Item - I and Item-II is:

Item-I	Item-II
(a) High density polythene	(I)Peroxide catalyst
(b) Polyacrylonitrile	(II) Condensation at hightemperature & pressure
(c) Novolac	(III) Zicgler-Natta catalyst
(d) Nylon 6	(IV) Acid or base catalyst

[April 10, 2019 (II)]

Options:

A. (a) → (I V), (b) → (I I), (c) → (I), (d) → (I I I)

B. (a) → (I I), (b) → (I V), (c) → (I), (d) → (I I I)

C. (a) → (I I I), (b) → (I), (c) → (I I), (d) → (I V)

D. (a) → (I I I), (b) → (I), (c) → (I V), (d) → (I I)

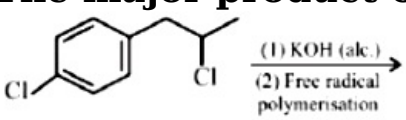
Answer: D

Solution:

Solution:
a. HDPE - Ziegler-Natta catalyst
b. Polyacrylonitrile - Peroxide catalyst
c. Novolac - Catalysed by acid or base
d. Nylon-6 - Condensation at high T and P

Question62

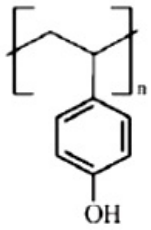
The major product of the following reaction is:



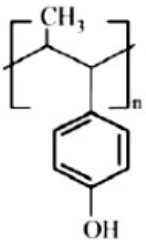
[April 9, 2019 (I)]

Options:

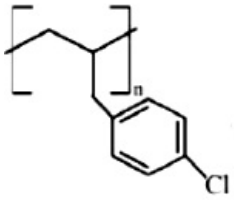
A.



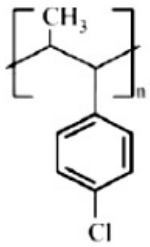
B.



C.



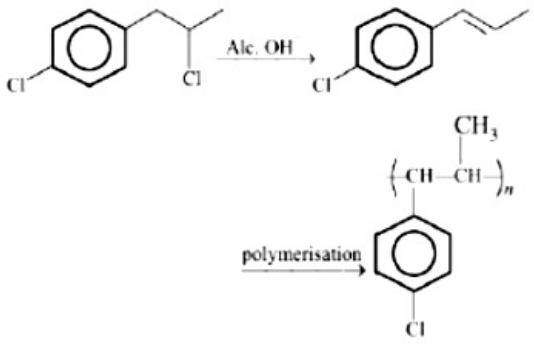
D.



Answer: D

Solution:

Solution:



Question63

Which of the following compounds is a constituent of the polymer

$[HN-\overset{\overset{O}{\|}}{C}-NH-CH_2]_n$?
[April 9, 2019 (II)]

Options:

A. N-Methyl urea

B. Formaldehyde

C. Methylamine

D. Ammonia

Answer: B

Solution:

The polymer $\left[NH-\overset{\overset{O}{\|}}{C}-NH-CH_2\right]_n$ is urea- formaldehyde resin.
Its monomers are urea and formaldehyde.

Question64

The copolymer formed by addition polymerization of styrene and acrylonitrile in the presence of peroxide is:
[Online April 15, 2018 (I)]

Options:

A. $\left[\begin{matrix}C_6H_5\\|\\C\\|\\CH_3\end{matrix}-\begin{matrix}CN\\|\\CH\end{matrix}-CH_2\right]_n$

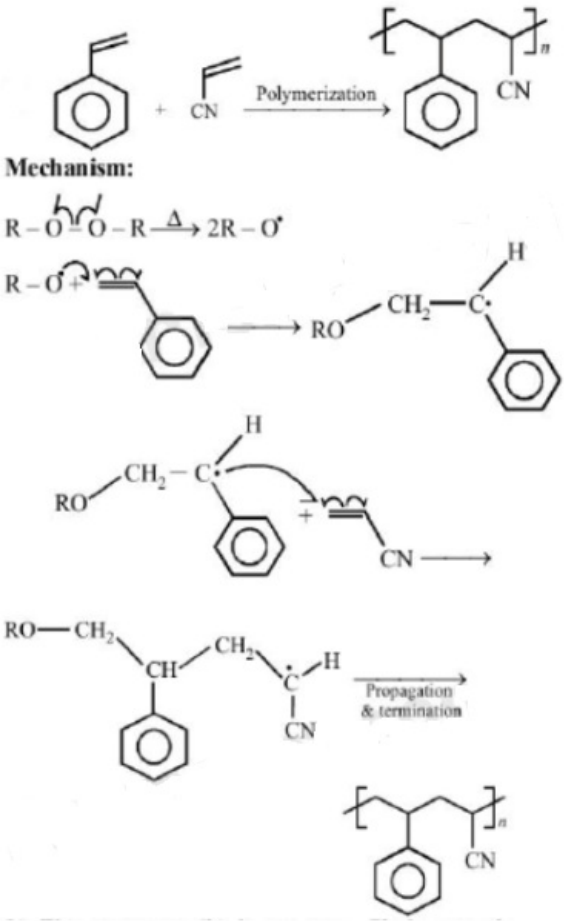
B. $\left[CH_2-\begin{matrix}\\|\\C\\|\\C_6H_5\end{matrix}H-CH_2-\begin{matrix}\\|\\CH\\|\\CN\end{matrix}\right]$

C. $\left[CH_2-\begin{matrix}C_6H_5\\|\\C\end{matrix}H\begin{matrix}CN\\|\\C\end{matrix}H-CH_2\right]_n$

D. $\left[\begin{matrix}\\|\\C\\|\\C_6H_5\end{matrix}H-CH_2-CH_2-\begin{matrix}CN\\|\\CH\end{matrix}\right]_n$

Answer: B

Solution:



Question65

Which of the following statements is not true? [Online April 15,2018 (II)]

Options:

A. Chain growth polymerisation involves homopolymerisation only

B. Chain growth polymerisation includes both homopolymerisation and copolymerisation

C. Nylon 6 is an example of step-growth polymerisation

D. Step growth polymerisation requires a bifunctional monomer

Answer: B

Solution:

Solution:
The statement (b) is not true. Chain growth polymerisation (or addition polymerisation) involves homopolymerisation only. Examples of such polymers include polythene, orlon and teflon.

Question66

The formation of which of the following polymers involves hydrolysis reaction?

[2017]

Options:

A. Nylon 6

B. Bakelite

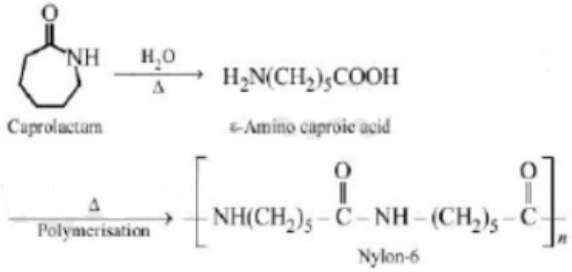
C. N yl on6, 6

D. Terylene

Answer: A

Solution:

Solution:
Formation of nylon-6 involves hydrolysis of caprolactam, (its monomer) in initial state.



Question67

Which of the following is a biodegradable polymer? [Online April 9,2017]

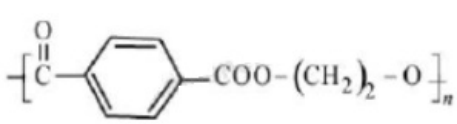
Options:

A. $\left[HN-(CH_2)_5\overset{O}{\overset{\|}{C}}-NH-CH_2-\overset{O}{\overset{\|}{C}} \right]_n$

B. $\left[HN-(CH_2)_5-\overset{O}{\overset{\|}{C}} \right]_n$

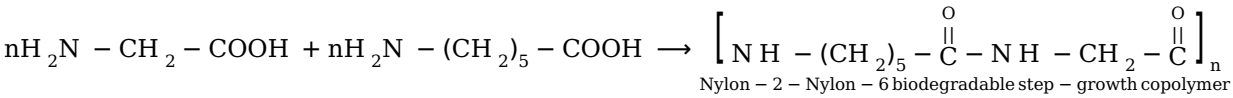
C. $\left[HN-(CH_2)_6NHCO-(CH_2)_4-\overset{O}{\overset{\|}{C}} \right]_n$

D.



Answer: A

Solution:



Nylon − 2 − Nylon − 6 biodegradable step − growth copolymer

Question68

Which of the following polymers is synthesized using a free radical polymerisation technique?
[Online April 10,2016]

Options:

A. Terylene

B. Melamine polymer

C. Nylon 6,6

D. Teflon

Answer: D

Solution:

Teflon or polytetrafluoroethene is an addition polymer and its monomer is alkene derivative ($CF_2 = CF_2$) whereas terylene, melamine and nylon-6, 6 are condensation polymers.

Question69

Which of the following statements about low density polythene is FALSE?
[2016]

Options:

A. Its synthesis requires dioxygen or a peroxide initiator as a catalyst.

B. It is used in the manufacture of buckets, dust-bins etc.

C. Its synthesis requires high pressure.

D. It is a poor conductor of electricity.

Answer: B

Solution:

Solution:
High density polythene is used in the manufacture of housewares like buckets, dustbins, bottles, pipes etc. Low density polythene is used for insulating electric wires and in the manufacture of flexible pipes, toys, coats, bottles etc.

Question70

Assertion : Rayon is a semisynthetic polymer whose properties are better than natural cotton.
Reason : Mechanical and aesthetic properties of cellulose can be improved by acetylation.
[Online April 9,2016]

Options:

A. Both assertion and reason are correct, but the reason is not the correct explantion for the assertion

B. Both assertion and reason are correct, and the reason is the correct explantion for the assertion

C. Assertion is incorrect statement, but the reason is correct.

D. Both assertion and reason are incorrect.

Answer: A

Question71

Which one of the following structures represents the neoprene polymer ?
[Online April 11, 2015]

Options:

A. $\left[\begin{array}{l} C \\ | \\ C_6H_5 \end{array} H - CH_2 \right]_n$

B. $\left[CH_2 - \begin{array}{l} C \\ | \\ CN \end{array} H \right]_n$

C. $\left[CH_2 - \begin{array}{l} C \\ | \\ CI \end{array} = CH - CH_2 \right]_n$

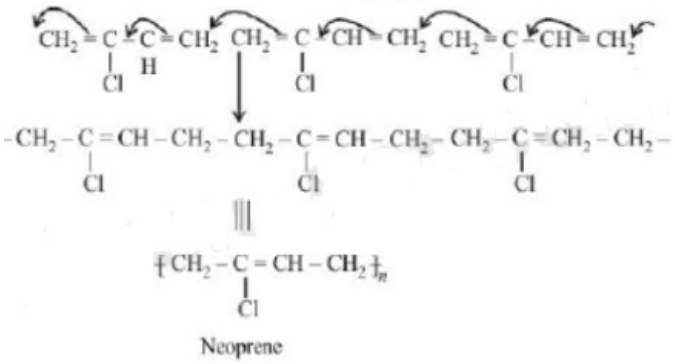
D. $\left[CH_2 - \begin{array}{l} C \\ | \\ CI \end{array} H \right]_n$

Answer: C

Solution:

Solution:
Chloroprene

$\left(CH_2 = CH - \overset{Cl}{\underset{|}{C}} = CH_2\right)$ is a monomer of neoprene.



Question72

Which polymer is used in the manufacture of paints and lacquers? [2015]

Options:

A. Polypropene

B. Polyvinyl chloride

C. Bakelite

D. Glyptal

Answer: D

Solution:

Solution:
Glyptal is used in the manufacture of paints and lacquers.

Question73

Match the polymers in column - A with their main uses in column - B and choose the correct answer:

Column - A	Column - B
(A) Polystrene	(i) Paints and lacquers
(B) Glyptal	(ii) Rain coats
(C) Polyvinyl chloride	(iii) Manufacture of toys
(D) Bakelite	(iv) Computer discs

[Online April 10,2015]

Options:

A. (A) − (ii), (B) − (i), (C) − (iii), (D) − (iv)

B. (A) − (ii), (B) − (iv), (C) − (iii), (D) − (i)

C. (A) − (iii), (B) − (iv), (C) − (ii), (D) − (i)

D. (A) − (iii), (B) − (i), (C) − (ii), (D) − (iv)

Answer: D

Solution:

Solution:

Polymer	Use
Polystyrene	Manufacture of toys
Glyptal	Paints and lacquers.
PV.C.	Rain Coats
Bakelite	Computer dises

Question74

Which one is classified as a condensation polymer? [2014]

Options:

A. Dacron

B. Neoprene

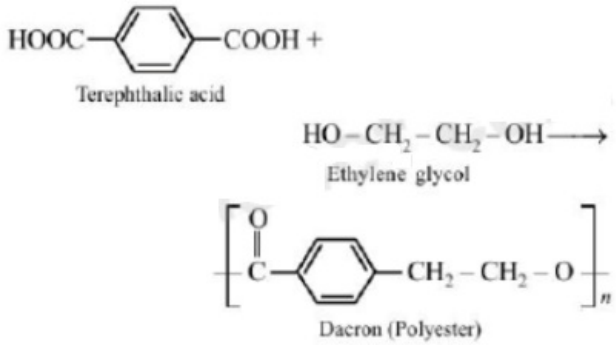
C. Teflon

D. Acrylonitrile

Answer: A

Solution:

Solution:
Except dacron (terylene) all are additive polymers. Terephthalic acid condenses with ethylene glycol to give dacron.



Question75

Which one of the following is an example of thermosetting polymers? [Online April 19,2014]

Options:

A. Neoprene

B. Buna-N

C. Nylon 6,6

D. Bakelite

Answer: D

Solution:

Solution:
Polymers which change irreversibly into hard and rigid material on heating are known as thermosetting polymers e.g bakelite.

Question76

Structure of some important polymers are given. Which one represents Buna-S? [Online April 9,2014]

Options:

A. $\left[CH_2 - \overset{CH_3}{\underset{|}{C}} = CH - CH_2 \right]_n$

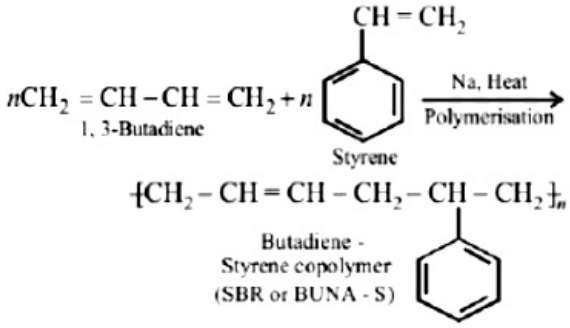
B. $\left[CH_2 - CH = CH - CH_2 - \underset{C_6H_5}{\underset{|}{C}}H - CH_2 \right]_n$

C. $\left[CH_2 - CH = CH - CH_2 - \underset{CN}{\underset{|}{C}}H - CH_2 \right]_n$

D. $\left[CH_2 - \overset{Cl}{\underset{|}{C}} = CH - CH_2 \right]_n$

Answer: B

Solution:



Question77

Which of the following polymer is a polyamide? [Online April 23, 2013]

Options:

A. Terylene

B. Nylon

C. Rubber

D. Vulcanised rubber

Answer: B

Solution:

Solution:
Nylon is a polyamide fibre. It is prepared by the condensation polymerisation of adipic acid (HOOC. $(CH_2)_4COOH$) and hexamethylene diamine (H_2N . $(CH_2)_6 \cdot NH_2$)

Question78

If a polythene sample contains two monodisperse fractions in the ratio 2 : 3 with degree of polymerization 100 and 200 , respectively, then its weight average molecular weight will be: [Online April 9, 2013]

Options:

A. 4900

B. 4600

C. 4300

D. 5200

Answer: A

Solution:

(a) Degree of polymerization of Ist polymer $= 100$
Degree of polymerization of IInd polymer $= 200$
$N_1 = 2; N_2 = 3$
Molar mass of monomer (ethylene) of polythene
$= 28g / mol$
Now,
Degree of polymerization is given as:

$$DP = \frac{\text{Molar mass of monomer}}{\text{Molar mass of polymer}}$$

Thus,
Molar mass of Ist polymer $(M_1) = 100 \times 28 = 2800g / mol$
Molar mass of IInd polymer $(M_2) = 200 \times 28 = 5600g / mol$
Now, the weight average molecular weight is given as:

$$M_w = \frac{\sum n_i M_i^2}{\sum n_i M_i}$$

$$\therefore M_w = \frac{n_1 M_1^2 + n_2 M_2^2}{n_1 M_1 + n_2 M_2} = \frac{2(2800)^2 + 3(5600)^2}{2 \times 2800 + 3 \times 5600}$$

$= 4900g / mol$

Question79

Bakelite is obtained from phenol by reacting with: [Online April 25,2013]

Options:

A. Acetal

B. CH_3CHO

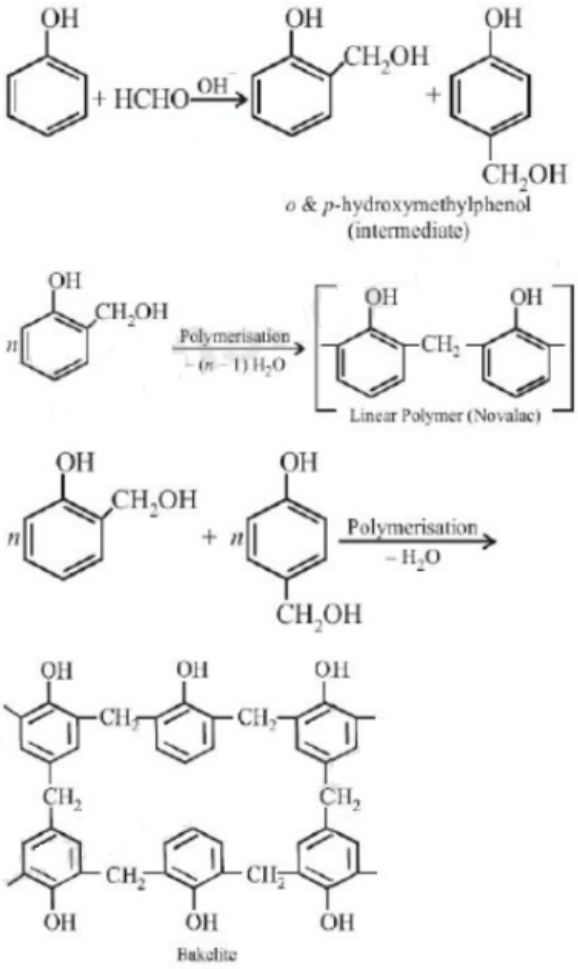
C. HCHO

D. Chlorobenzene

Answer: C

Solution:

Phenol and formaldehyde undergo condensation polymerisation under two different conditions to give a cross linked polymer called bakelite.



Question80

The polymer used for optical lenses is : [Online April 22, 2013]

Options:

A. polypropylene

B. polyvinyl chloride

C. polythene

D. polymethyl methacrylate

Answer: D

Solution:

Solution:
Polymethyl methacrylate is hard, fairlyrigid. It is used for optical lenses.

Question81

Which of the following is a polyamide?

[Online May 19,2012, April 23,2013]

Options:

A. Teflon

B. Orlon

C. Nylon

D. Terylene

Answer: C

Solution:

Solution:
Nylon is an aliphatic polyamide.

Question82

Which one of the following is a chain growth polymerisation? [Online May 7,2012]

Options:

A. Nucleic acid

B. Polystyrene

C. Protein

D. Starch

Answer: B

Solution:

Solution:
Polystyrene is a chain growth polymer.

Question83

The species which can best serve as an initiator for the cationic polymerisation is: [2012]

Options:

A. $LiAlH_4$

B. HNO_3

C. $AlCl_2$

D. BaLi

Answer: C

Solution:

Solution:
Lewis acids are the most common compounds used for initiation of cationic polymerisation. The more popular Lewis acids are $SnCl_4$, $AlCl_3$, BF_3 and $TiCl_4$

Question84

Synthetic polymer bakelite can be prepared from following compounds [Online May 12, 2012]

Options:

A. Styrene and vinyl chloride

B. Acrylonitrile and vinyl chloride

C. Adipic acid and ethylene glycol

D. Phenol and formaldehyde

Answer: D

Solution:

Solution:
Bakelite is prepared condensation of phenol and formaldehyde.

Question85

Bakelite is obtained from phenol by reacting it with [Online May 26, 2012]

Options:

A. Acetaldehyde

B. Chlorobenzene

C. Formaldehyde

D. Acetamide

Answer: C

.

Question86

Thermosetting polymer, bakelite is formed by the reaction of phenol with :
[2011RS]

Options:

A. CH_3CHO

B. HCHO

C. HCOOH

D. CH_3CH_2CHO

Answer: B

Question87

The polymer containing strong intermolecular forces e.g. hydrogen bonding, is
[2010]

Options:

A. teflon

B. nylon 6,6

C. polystyrene

D. natural rubber

Answer: B

Solution:

Solution:
Nylon 6,6 has amide linkage capable of forming hydrogen bonding.

Question88

Buna-N synthetic rubber is a copolymer of:

[2009]

Options:

A. $H_2C = CH - CH = CH_2$ and $H_5C_6 - CH = CH_2$

B. $H_2C = CH - CN$ and $H_2C = CH - CH = CH_2$

C. $H_2C = CH - CN$ and $H_2C = CH - \underset{\underset{CH_3}{|}}{C} = CH_2$

D. $H_2C = CH - \overset{\overset{CI}{|}}{C} = CH_2$ and $H_2C = CH - CH = CH_2$

Answer: B

Solution:

Solution:
Buna $-N$ is a copolymer of butadiene
($CH_2 = CH - CH = CH_2$) and acrylonitrile ($CH_2 = CH\,CN$)

Question89

Bakelite is obtained from phenol by reacting with [2008]

Options:

A. $(CH_2OH)_2$

B. $CH_3CH\,O$

C. CH_3COCH_3

D. HCHO

Answer: D

Question90

Which of the following is a polyamide? [2005]

Options:

A. Bakelite

B. Terylene

C. Nylon-66

D. Teflon

Answer: C

Solution:

Solution:
Nylon is a general name for all synthetic fibres forming polyamides.

Question91

Which of the following is fully fluorinated polymer? [2005]

Options:

A. PVC

B. Thiokol

C. Teflon

D. Neoprene

Answer: C

Solution:

Solution:
Teflon is polymer of $CF_2 = CF_2$

Question92

Nylon threads are made of [2003]

Options:

A. polyester polymer

B. polyamide polymer

C. polyethylene polymer

D. polyvinyl polymer

Answer: B

Solution:

Nylon is a polyamide polymer.

Question93

Polymer formation from monomers starts by [2002]

Options:

A. condensation reaction between monomers

B. coordinate reaction between monomers

C. conversion of monomer to monomer ions by protons

D. hydrolysis of monomers.

Answer: A

Solution:

Polymerisation starts either by condensation or addition reactions between monomers. Condensation polymers are formed by the combination of monomers with the elimination of simple molecules. Whereas addition polymers are formed by the addition of the molecules of the monomer or monomers to form a large molecule without elimination of any thing.
